

PROJECT PROPOSAL

BBR: Congestion-Based Congestion Control — Reproduction & Analysis

Instructor: Prof. Minsung Kim | Members: Mihir Kulkarni • Chaitanya Ranaware • Piyoosha Gadi

Paper Background

BBR: Congestion-Based Congestion Control was authored by Neal Cardwell, Yuchung Cheng, C. Stephen Gunn, Soheil Hassas Yeganeh, and Van Jacobson — all engineers and researchers at Google — and published in *ACM Queue* (2016), subsequently republished in *ACM SIGCOMM Computer Communication Review*. The paper introduces a fundamentally different approach to TCP congestion control. Traditional algorithms such as CUBIC and RENO treat packet loss as the primary indicator of network congestion. This assumption breaks down in modern networks where large router buffers cause packets to queue extensively before being dropped, a phenomenon known as buffer bloat — resulting in artificially inflated RTTs and underutilized bandwidth despite no actual congestion.

BBR addresses this by directly modelling the network path. Rather than reacting to loss events, it continuously estimates two key physical properties: the bottleneck bandwidth (BtlBw) — the maximum rate the slowest link on the path can sustain — and the minimum round-trip propagation time (RTprop). Together these define the Bandwidth-Delay Product (BDP), which BBR uses to set the congestion window and pacing rate optimally. The algorithm cycles through four distinct phases: STARTUP (exponential ramp-up to discover BtlBw), DRAIN (reducing the queue built during startup), PROBE_BW (steady-state operation with periodic bandwidth probing), and PROBE_RTT (brief throughput reduction to refresh the RTprop estimate). This design allows BBR to maintain high throughput without inflating the queue, achieving both low latency and high link utilization simultaneously.

The real-world impact of BBR is substantial. It has been deployed in the Linux kernel since version 4.9, adopted by Google across YouTube, google.com, and Google Cloud, and integrated into the QUIC transport protocol underlying HTTP/3. In Google's own measurements, BBR achieved throughput gains of up to 2,700× over CUBIC on high-latency, lossy links — making this one of the most consequential networking papers of the past decade.

Project Objectives

The goal of this project is to reproduce the core experimental results of the BBR paper and extend the analysis to scenarios not fully explored in the original work. Specifically, we aim to:

- Implement a controlled testbed using Linux virtual machines with `tc netem` for link emulation and `iperf3` for traffic generation, replicating the paper's experimental setup as closely as possible.
- Reproduce the paper's throughput and latency comparisons between BBR, CUBIC, and RENO under varying bottleneck bandwidths, buffer sizes, and propagation delays — validating our results against the published figures.
- Investigate buffer bloat behavior: demonstrate how CUBIC's queue-filling tendency inflates RTT while BBR maintains near-minimum latency at equivalent throughput levels.
- Analyze inter-protocol fairness when BBR and CUBIC flows compete on a shared bottleneck, examining whether BBR's pacing model causes it to starve or be starved by loss-based flows.
- Present the algorithmic design clearly, connecting BBR's pacing and windowing mechanisms to the theoretical concepts covered in CS 552, including the TCP/IP model, transport layer design, and congestion control theory.

Primary Reference

Cardwell, N., Cheng, Y., Gunn, C. S., Hassas Yeganeh, S., & Jacobson, V. (2016). BBR: Congestion-based congestion control. *ACM Queue*, 14(5). Republished in *ACM SIGCOMM Computer Communication Review*, 47(2), 92–101.